

ICMP (Internet Control Message protocol)

Main Uses:-

- **Error Reporting:** If a data packet cannot reach its destination.
- **Network Diagnostics:** It is the foundation for common troubleshooting tools:
 - Ping: Uses "Echo Request" and "Echo Reply" messages to check if a device is reachable and to measure latency.
 - Traceroute: Uses "Time Exceeded" messages to map the physical path (hops) a packet takes across routers to its destination.
- **Path Optimization:** Routers use ICMP Redirect messages to inform hosts about more efficient routes for their data.

Common types/Formats :-

- *Type 0 - Echo reply*
- *Type 3 - Destination unreachable*
- *Type 5 - Redirect Message*
- *Type 8 - Echo Request*
- *Type 11 - Time Exceeded*
- *Type 12 - Parameter problem*

source:- [Internet Control Message Protocol \(ICMP\) - GeeksforGeeks](#)

Attacks using ICMP :-

- ICMP flood or DDoS
- Ping of Death: Sending oversized ICMP packets to crash older networking equipment. (Not very relevant now-a-days. It occurs older equipment, and networked printers produced in the era(before 2000's) often lacked the logic to handle malformed, fragmented packets that exceeded the 65,535-byte limit.)
- ICMP tunneling: Hiding malicious traffic within standard ICMP packets to bypass firewalls.

Most Common Protection

Limiting ICMP traffic at the firewall level.

Available Datasets

- **SDN ICMP Flood Dataset**

<https://data.mendeley.com/datasets/zdczzpkd8z/2>

[Research paper link](#)

- **ARP Spoofing, SYN, PING flood attacks**

[Kaggle dataset](#)

But these are directly wireshark packets and have not been converted into flow features

[Research Paper](#)